SQL

CHALLENGES

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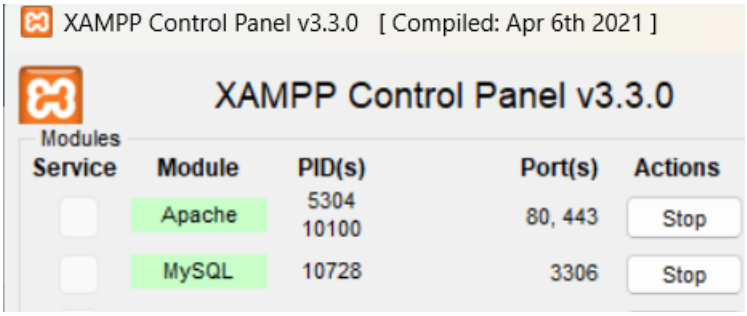
Avant de commancer les challenges il faut :

1. Activer le serveur mysql



2. Ouvrir le terminal dans XAMPP et se connecter avec MYSQL

```
hp@DESKTOP-GGGBH99 c:\Users\hp\h p\Bureau\xamp
# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
```

3. Créer une database :

```
MariaDB [(none)]> create database Challenges_sql;
Query OK, 1 row affected (0.003 sec)
```

4.Utiliser cette database :

```
MariaDB [(none)]> use Challenges_sql;
Database changed
```

Challenge #1: The Architect's Schema

- **Challenge 1: Write a SQL script to create a table named `library_books` with exactly these specifications:**
  - **`id`: A unique number that increments automatically.**
  - **`title`: Text up to 150 chars (cannot be empty).**
  - **`author`: Text up to 100 chars.**
  - **`published_year`: A 4-digit year.**
  - **`status`: Either 'Available', 'Borrowed', or 'Lost'.**
  - **`price`: A decimal number (e.g., 19.99).**

```
MariaDB [Challenges_sql]> create table library_books(
    -> id int auto_increment primary key,
    -> title varchar(150) not null,
    -> author varchar(100),
    -> published_year date,
    -> status enum('Available', 'Borrowed','Lost'),
    -> price decimal(5,2)
    -> );
Query OK, 0 rows affected (0.019 sec)
```

NB :

« Pensez à remplacer le type `DATE` par `YEAR` pour le champ `published_year`. Il faut que l'on puisse juste entrer l'année comme 2005 ou 1990. »

Challenge #2: Data Seeding

Challenge 2: Data Seeding Insert 8 specific books into your table. Ensure at least:

- One book is from the year 1950.
- Two books are by the same author.
- One book has a `NULL` price (representing a donated book)

```
MariaDB [Challenges_sql]> INSERT INTO library_books (title, author, published_year, status, price)
    -> VALUES
    -> ('The Old Man and the Sea','Ernest Hemingway',1950,'Available',12.99),
    -> ('Clean Code','Robert Martin',2008,'Borrowed',29.99),
    -> ('Clean Architecture','Robert Martin',2017,'Available',31.50),
    -> ('The Pragmatic Programmer','Andrew Hunt',1999,'Available',25.50),
    -> ('Introduction to Algorithms','Thomas H. Cormen',2009,'Borrowed',45.75),
    -> ('Donated Book','Unknown',2015,'Available',NULL),
    -> ('Design Patterns','Erich Gamma',1994,'Lost',39.99),
    -> ('Learning SQL','Alan Beaulieu',2020,'Available',25.50);
Query OK, 8 rows affected, 8 warnings (0.009 sec)
Records: 8  Duplicates: 0  Warnings: 8
```

Challenge #3:

- **CHALLENGE 3 :WRITE SINGLE SQL QUERIES FOR EACH OF THE FOLLOWING:**

1. **The Catalog:** List all books, but only show the `title`, `author`, and `price`.
2. **The Budget Filter:** Find all books that cost between **10 MAD** and **20 MAD**.
3. **The Modernist:** Find all books published after the year **2010**.
4. **The Search Bar:** Find all books that have the word **"SQL"** anywhere in their title.
5. **The Availability Check:** List all books that are NOT currently 'Lost', sorted by the most recent `published_year`.
6. **The Author Audit:** List all unique (`DISTINCT`) authors in the library.
7. **The Formatting:** Show the book titles in **UPPERCASE** and the price rounded to the nearest whole number.

1. The Catalog:

```
MariaDB [Challenges_sql]> select title,author,price from library_books;
+----------------------------+------------------+-------+
| title                      | author           | price |
+----------------------------+------------------+-------+
| The Old Man and the Sea    | Ernest Hemingway | 12.99 |
| Clean Code                 | Robert Martin    | 29.99 |
| Clean Architecture         | Robert Martin    | 31.50 |
| The Pragmatic Programmer   | Andrew Hunt      | 25.50 |
| Introduction to Algorithms | Thomas H. Cormen | 45.75 |
| Donated Book               | Unknown          |  NULL |
| Design Patterns            | Erich Gamma      | 39.99 |
| Learning SQL               | Alan Beaulieu    | 25.50 |
+----------------------------+------------------+-------+
8 rows in set (0.001 sec)
```

2.The Budget Filter:

```
MariaDB [Challenges_sql]> select title from library_books where price between 10 and 20;
+-------------------------+
| title                   |
+-------------------------+
| The Old Man and the Sea |
+-------------------------+
1 row in set (0.001 sec)
```

3. The Modernist :

```
MariaDB [Challenges_sql]> select *from library_books where published_year >2010;
+----+--------------------+---------------+----------------+-----------+-------+
| id | title              | author        | published_year | status    | price |
+----+--------------------+---------------+----------------+-----------+-------+
|  3 | Clean Architecture | Robert Martin |           2017 | Available | 31.50 |
|  6 | Donated Book       | Unknown       |           2015 | Available |  NULL |
|  8 | Learning SQL       | Alan Beaulieu |           2020 | Available | 25.50 |
+----+--------------------+---------------+----------------+-----------+-------+
3 rows in set (0.001 sec)
```

4. The Search Bar:

```
MariaDB [Challenges_sql]> select *from library_books where title like "%sql" ;
+----+--------------+---------------+----------------+-----------+-------+
| id | title        | author        | published_year | status    | price |
+----+--------------+---------------+----------------+-----------+-------+
|  8 | Learning SQL | Alan Beaulieu |           2020 | Available | 25.50 |
+----+--------------+---------------+----------------+-----------+-------+
1 row in set (0.001 sec)
```

5. **The Availability Check:**

```
MariaDB [Challenges_sql]> select * from library_books where status != 'Lost' order by published_year desc;
+----+---------------------------+------------------+----------------+-----------+-------+
| id | title                     | author           | published_year | status    | price |
+----+---------------------------+------------------+----------------+-----------+-------+
|  8 | Learning SQL              | Alan Beaulieu    |           2020 | Available | 25.50 |
|  3 | Clean Architecture        | Robert Martin    |           2017 | Available | 31.50 |
|  6 | Donated Book              | Unknown          |           2015 | Available |  NULL |
|  5 | Introduction to Algorithms | Thomas H. Cormen |           2009 | Borrowed  | 45.75 |
|  2 | Clean Code                | Robert Martin    |           2008 | Borrowed  | 29.99 |
|  4 | The Pragmatic Programmer  | Andrew Hunt      |           1999 | Available | 25.50 |
|  1 | The Old Man and the Sea   | Ernest Hemingway |           1950 | Available | 12.99 |
+----+---------------------------+------------------+----------------+-----------+-------+
7 rows in set (0.001 sec)
```

6. **The Author Audit:**

```
MariaDB [Challenges_sql]> select distinct  author from library_books;
+------------------+
| author           |
+------------------+
| Ernest Hemingway |
| Robert Martin    |
| Andrew Hunt      |
| Thomas H. Cormen |
| Unknown          |
| Erich Gamma      |
| Alan Beaulieu    |
+------------------+
7 rows in set (0.001 sec)
```

7 **The Formatting:**

```
MariaDB [Challenges_sql]> select upper(title),round(price) from library_books;
+---------------------------+--------------+
| upper(title)              | round(price) |
+---------------------------+--------------+
| THE OLD MAN AND THE SEA   |           13 |
| CLEAN CODE                |           30 |
| CLEAN ARCHITECTURE        |           32 |
| THE PRAGMATIC PROGRAMMER  |           26 |
| INTRODUCTION TO ALGORITHMS |           46 |
| DONATED BOOK              |         NULL |
| DESIGN PATTERNS           |           40 |
| LEARNING SQL              |           26 |
+---------------------------+--------------+
8 rows in set (0.001 sec)
```

Challenge #4: INNER JOIN

1. **Setup:** Create a second table called `categories` with `id` (PK) and `name`.
2. **link:** In your `library_books` table, add a column `category_id` (INT).
3. **Action:** Assign a category ID to each book in your database.
4. **The Query:** Write a `SELECT` statement that joins both tables to show the **Book Title** and the **Category Name** side-by-side

1.setup

```
MariaDB [Challenges_sql]> create table categories(
    -> id int auto_increment primary key,
    -> name varchar(50) not null
    -> );
Query OK, 0 rows affected (0.013 sec)
```

2.link

```
MariaDB [Challenges_sql]> ALTER TABLE library_books
    -> ADD category_id INT,
    -> ADD FOREIGN KEY (category_id) REFERENCES categories(id);
Query OK, 8 rows affected (0.064 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Après on va insérer des données au table categories

```
MariaDB [Challenges_sql]> INSERT INTO categories (name)
    -> VALUES
    -> ('Novel'),
    -> ('Programming'),
    -> ('Algorithms'),
    -> ('Database');
Query OK, 4 rows affected (0.005 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

Après on va insérer les données aussi au colonne category_id dans la table library_books

```
MariaDB [Challenges_sql]> update library_books
    -> set category_id = case id
    -> when 1 then 1
    -> when 2 then 2
    -> when 3 then 2
    -> when 4 then 2
    -> when 5 then 3
    -> when 6 then 2
    -> when 7 then 2
    -> when 8 then 4
    -> end
    -> where id in (1,2,3,4,5,6,7,8);
Query OK, 8 rows affected (0.009 sec)
Rows matched: 8  Changed: 8  Warnings: 0
```

Le tableau après insertion

```
MariaDB [Challenges_sql]> select *from library_books;
+----+----------------------------+------------------+----------------+-----------+-------+-------------+
| id | title                      | author           | published_year | status    | price | category_id |
+----+----------------------------+------------------+----------------+-----------+-------+-------------+
|  1 | The Old Man and the Sea    | Ernest Hemingway |           1950 | Available | 12.99 |           1 |
|  2 | Clean Code                 | Robert Martin    |           2008 | Borrowed  | 29.99 |           2 |
|  3 | Clean Architecture         | Robert Martin    |           2017 | Available | 31.50 |           2 |
|  4 | The Pragmatic Programmer   | Andrew Hunt      |           1999 | Available | 25.50 |           2 |
|  5 | Introduction to Algorithms | Thomas H. Cormen |           2009 | Borrowed  | 45.75 |           3 |
|  6 | Donated Book               | Unknown          |           2015 | Available |  NULL |           2 |
|  7 | Design Patterns            | Erich Gamma      |           1994 | Lost      | 39.99 |           2 |
|  8 | Learning SQL               | Alan Beaulieu    |           2020 | Available | 25.50 |           4 |
+----+----------------------------+------------------+----------------+-----------+-------+-------------+
8 rows in set (0.001 sec)
```

4.La requête SELECT

```
MariaDB [Challenges_sql]> select title , name
    -> from library_books
    -> join categories
    -> on library_books.category_id = categories.id;
+----------------------------+-------------+
| title                      | name        |
+----------------------------+-------------+
| The Old Man and the Sea    | Novel       |
| Clean Code                 | Programming |
| Clean Architecture         | Programming |
| The Pragmatic Programmer   | Programming |
| Introduction to Algorithms | Algorithms  |
| Donated Book               | Programming |
| Design Patterns            | Programming |
| Learning SQL               | Database    |
+----------------------------+-------------+
8 rows in set (0.001 sec)
```

Challenge #5: AggregatesBrief

Target: Perform calculations directly in the database. Write single queries to calculate:

1. **Total Inventory Value:** The sum of all book prices.
2. **Stock Count:** The total number of books currently marked as 'Available'.
3. **Price Extremes:** The most expensive and the cheapest book in one query.
4. **Average Cost:** The average price of all books in your library.

1. **Total Inventory Value:**

```
MariaDB [Challenges_sql]> select sum(price) as The_sum_of_books_price from library_books;
+-------------------------+
| The_sum_of_books_price |
+-------------------------+
|                  211.22 |
+-------------------------+
1 row in set (0.001 sec)
```

2. **Stock Count:**

```
MariaDB [Challenges_sql]> select count(status) as The_sum_of_books_Available from library_books where status  like "%Available%";
+----------------------------+
| The_sum_of_books_Available |
+----------------------------+
|                          5 |
+----------------------------+
1 row in set (0.001 sec)
```

3. **Price Extremes:**

```
MariaDB [Challenges_sql]> (select *  from library_books order by price desc limit 1) UNION (select *  from library_books order by price asc limit 1
+----+---------------------------+-----------------+----------------+-----------+-------+-------------+
| id | title                     | author          | published_year | status    | price | category_id |
+----+---------------------------+-----------------+----------------+-----------+-------+-------------+
|  5 | Introduction to Algorithms | Thomas H. Cormen |           2009 | Borrowed  | 45.75 |           3 |
|  6 | Donated Book              | Unknown         |           2015 | Available |  NULL |           2 |
+----+---------------------------+-----------------+----------------+-----------+-------+-------------+
2 rows in set (0.002 sec)
```

4. average price

```
MariaDB [Challenges_sql]> select avg(price) as average_price from library_books;
+----------------+
| average_price |
+----------------+
|      30.174286 |
+----------------+
1 row in set (0.001 sec)
```

Challenge #6: GROUP BY & HAVING

1. **Status Report:** Show each `status` (Available, Borrowed, Lost) and how many books belong to each.
2. **Author Performance:** Show the total value of books owned by each `author`.
3. **The Filter:** Show only the authors whose total book value is **greater than 30 MAD** (use `HAVING`).

1. **Status Report:**

```
MariaDB [Challenges_sql]> select status,count(*) as le_nombre_des_elements
    -> from library_books
    -> group by status;
+-----------+------------------------+
| status    | le_nombre_des_elements |
+-----------+------------------------+
| Available |                      5 |
| Borrowed  |                      2 |
| Lost      |                      1 |
+-----------+------------------------+
3 rows in set (0.001 sec)
```

2. **Author Performance:**

```
MariaDB [Challenges_sql]> select author ,sum(price) as sum_value
    -> from library_books
    -> group by author;
+------------------+-----------+
| author           | sum_value |
+------------------+-----------+
| Alan Beaulieu    |     25.50 |
| Andrew Hunt      |     25.50 |
| Erich Gamma      |     39.99 |
| Ernest Hemingway |     12.99 |
| Robert Martin    |     61.49 |
| Thomas H. Cormen |     45.75 |
| Unknown          |      NULL |
+------------------+-----------+
7 rows in set (0.001 sec)
```

3. **The Filter:**

```
MariaDB [Challenges_sql]> select author ,sum(price) as sum_value
    -> from library_books
    -> group by author
    -> having sum(price) > 30;
+------------------+-----------+
| author           | sum_value |
+------------------+-----------+
| Erich Gamma      |     39.99 |
| Robert Martin    |     61.49 |
| Thomas H. Cormen |     45.75 |
+------------------+-----------+
3 rows in set (0.001 sec)
```